Conditioning, Guidance, and Memorization in Diffusion Image Generation

Aathi Muthu
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Abstract

This project studies class conditioning and classifier-free guidance using pretrained EDM models on CIFAR-10, smaller EDM networks trained on restricted subsets, and a conditional U-Net trained from scratch on MNIST. Paired noise allows controlled comparisons of guidance strength. Evaluation combines requested-class accuracy, feature FID, denoising trajectories, and nearest-training-image comparisons. Moderate guidance improves class matching, while stronger guidance can worsen FID. The 100-image CIFAR model produces close training-image matches, showing that falling training loss need not imply generalization.

1 Introduction

1.1 Problem and motivation

A class-conditional image generator should satisfy a requested class while preserving realistic variation. We examine when class structure emerges during denoising, how guidance changes accuracy and distributional similarity, and whether small training sets lead to memorization.

Karras et al.'s *Elucidating the Design Space of Diffusion-Based Generative Models* (EDM) [1] separates sampling, network preconditioning, and training choices. Its Heun sampler refines Euler proposals, while noise-dependent preconditioning stabilizes the denoiser's input and output scales. We use the supplied labs to study sampling with fixed pretrained networks and training with restricted data.

1.2 Related work

Denoising diffusion probabilistic models generate images through iterative denoising [2]. Score-based models describe generation using stochastic differential equations and an associated ODE [3]. EDM organizes and improves these design choices [1].

Classifier-free guidance [4] combines two denoiser estimates:

$$D_w(x;\sigma,c) = D_u(x;\sigma) + w\big(D_c(x;\sigma,c) - D_u(x;\sigma)\big).$$

Thus $w = 0$ ignores the request and $w = 1$ uses the conditional estimate. The pretrained experiments use separate conditional and unconditional EDM networks. Our trained models use 10% label dropout so one network learns both estimates. Frozen classifiers evaluate the samples.

1.3 Data

CIFAR-10 has 50,000 training and 10,000 test images: 32-by-32 RGB images in ten classes. Restricted-data experiments use balanced subsets of 100 and 1,000 training images. Test images provide the nearest-training-image baseline.

MNIST has 60,000 training and 10,000 test digits: 28-by-28 grayscale images scaled to $[-1, 1]$. We train on all 60,000 digits from random initialization. The frozen lab classifier provides labels and 128-dimensional features; all 10,000 test digits form the feature-FID reference.

2 Experimental protocol

Pretrained CIFAR sampling uses 18 deterministic Heun steps and 1,000 balanced requests per weight at $w = 0, 1, 1.5, 2, 3, 4$, with paired noise. Global Inception FID uses NVIDIA's 50,000-image training reference. Class FID averages ten comparisons of 100 generated images against 5,000 real images per class; it is omitted at $w = 0$. Accuracy measures evaluator agreement, while global FID ignores the requested class. These small-sample FIDs are not directly comparable with EDM's published 50,000-sample scores.

Each restricted-data run uses the lab's 7M-parameter EDM network, EDMLoss, 10% label dropout, Adam at $2 \cdot 10^{-4}$, batch 64, and EMA. Both complete 20,000 updates, with 1,000 unconditional samples evaluated every 4,000 updates. Nearest-image comparisons use 100 generated and 100 real test images.

MNIST uses the lab U-Net with 64 base channels, null label 10, Adam at $2 \cdot 10^{-4}$, batch 128, EMA, and 10,000 updates. Training minimizes clean-image MSE at uniformly sampled times on the lab's cosine path. Evaluation uses 32 DDIM steps and 1,000 balanced requests at $w = 0, 1, 2, 4$, with paired noise. Its classifier-feature FID has a separate scale from CIFAR Inception FID.

3 Preliminary results

Trajectories. In the three unconditional examples, recognizable content emerges between $\sigma \approx 3.3$ and 1.1; composition is mostly stable by $\sigma \approx 0.3$. For paired cat trajectories, both $w = 1$ and $w = 4$ show cat structure by $\sigma \approx 1.1$. The second $w = 4$ example shows clearer ears and a face already at $\sigma \approx 3.3$. Both change little in composition below $\sigma \approx 0.3$, but $w = 4$ strengthens contrast and facial features and changes some poses or proportions. These thresholds describe the displayed examples.

Pretrained CIFAR guidance. Global FID is nearly identical at $w = 0$ and $w = 1$ (30.97/30.93), although requested-class accuracy differs sharply (11.0%/95.8%). I favor $w = 1.5$: accuracy reaches 98.6%, global FID is 30.96, and class FID is lowest at 107.16. Weight two gives the highest accuracy, 99.5%, at global FID 32.22. Larger weights worsen both FIDs without improving accuracy. Visually, stronger guidance sharpens several cats, but the sixth column remains dark and ambiguous. These single-seed comparisons do not establish significance for small score differences.

Restricted-data training. For 100 training images, FID rises from 73.04 to 106.83 while the 200-update mean loss before the first and last snapshots falls from 0.0367 to 0.0179. Generated/test median nearest-training-image distances are 1.19/23.67; close object and background matches support memorization. For 1,000 images, FID rises from 92.00 to 132.47, loss falls from 0.1788 to

0.0694, and distance medians are 13.86/21.25. Many outputs are blurry partial matches, so exact-copy evidence is weaker. Lower pixel distances can also reflect blurred averages; these fixed-budget runs do not show that more data harms diffusion models.

The trained-model CFG grids score 100 images per weight. At $w = 0, 1, 2, 4$, accuracies are 7%, 96%, 100%, 100% for the 100-image model and 10%, 47%, 74%, 84% for the 1,000-image model. Guidance improves class matching, but visible artifacts remain.

MNIST guidance. Accuracy rises from 10.4% at $w = 0$ to 96.4% at $w = 1$ and 100% at $w = 2, 4$. Feature FID is lowest at $w = 1$ (22.10), then rises to 85.14 and 182.57. I favor $w = 1$ for distributional similarity or $w = 2$ for class adherence. Weight four adds no measured accuracy gain. The requested seven is clearer at higher weights, but classifier decisions alone do not establish perceptual quality.

3.1 Pretrained CIFAR-10 guidance

1,000 generated images per setting, balanced requests. Class FID uses 100 generated images per class and is omitted for $w = 0$.

w	Accuracy	FID	Class FID
0	0.110	30.97	–
1	0.958	30.93	111.40
1.5	0.986	30.96	107.16
2	0.995	32.22	107.87
3	0.991	35.59	112.81
4	0.990	38.22	116.79

3.2 CIFAR-10 training: 100 images

20,000 updates; 1,000 unconditional samples per snapshot.

Update	FID
4000	73.04
8000	88.55
12000	98.54
16000	105.63
20000	106.83

3.3 CIFAR-10 training: 1000 images

20,000 updates; 1,000 unconditional samples per snapshot.

Update	FID
4000	92.00
8000	111.34
12000	121.21
16000	129.61
20000	132.47

3.4 MNIST guidance

10,000 training updates; 1,000 samples per setting. Feature FID uses 10,000 real test images.

w	Accuracy	Feature FID
0	0.104	106.40
1	0.964	22.10
2	1.000	85.14
4	1.000	182.57

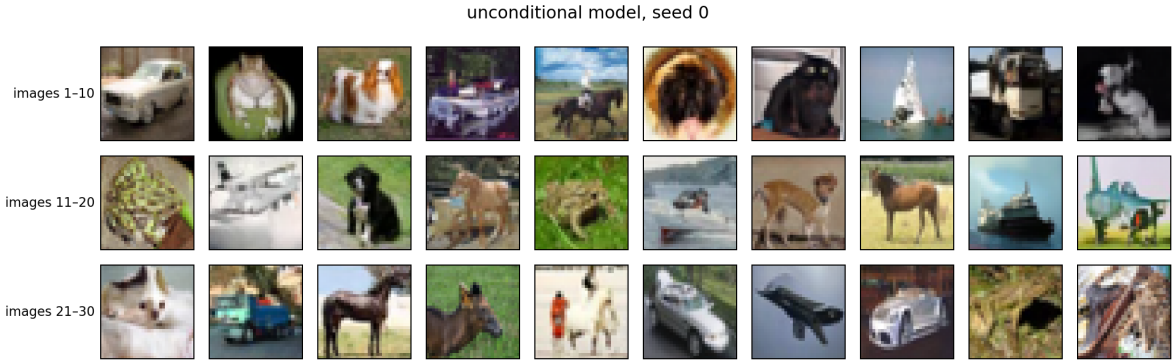


Figure 1: Unconditional samples show varied object classes without a class request.



Figure 2: Changing the class changes object identity while some colors and layouts persist within paired-noise columns.

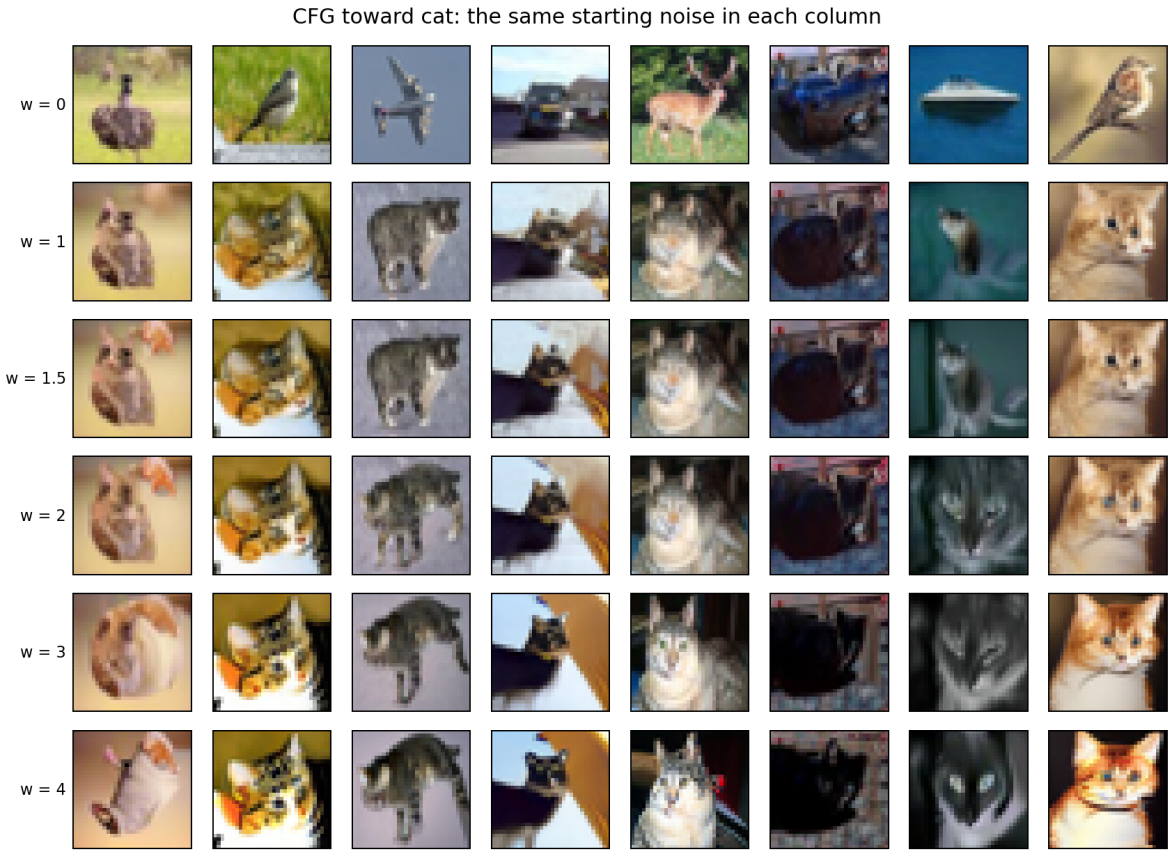


Figure 3: Stronger cat guidance sharpens several faces, but the dark sixth-column sample remains ambiguous.

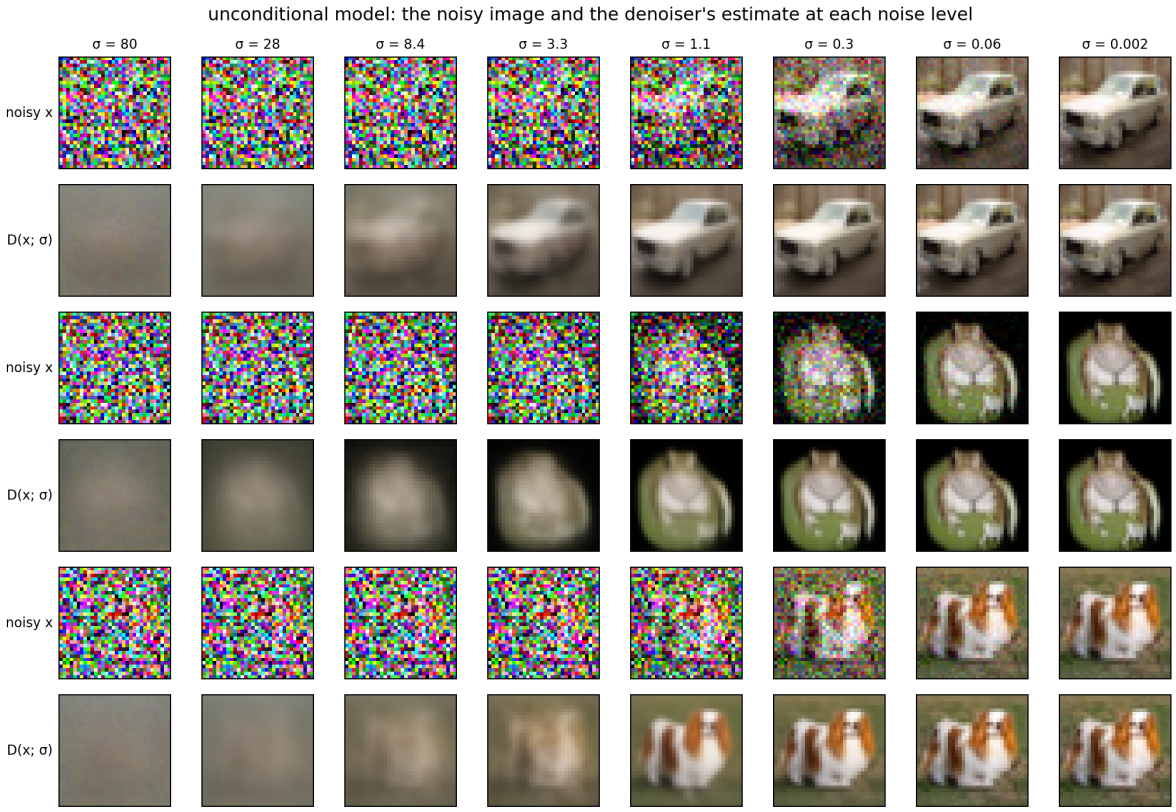


Figure 4: Recognizable content emerges at intermediate noise; late steps mainly sharpen detail. Upper rows show noisy states and lower rows show denoiser estimates.

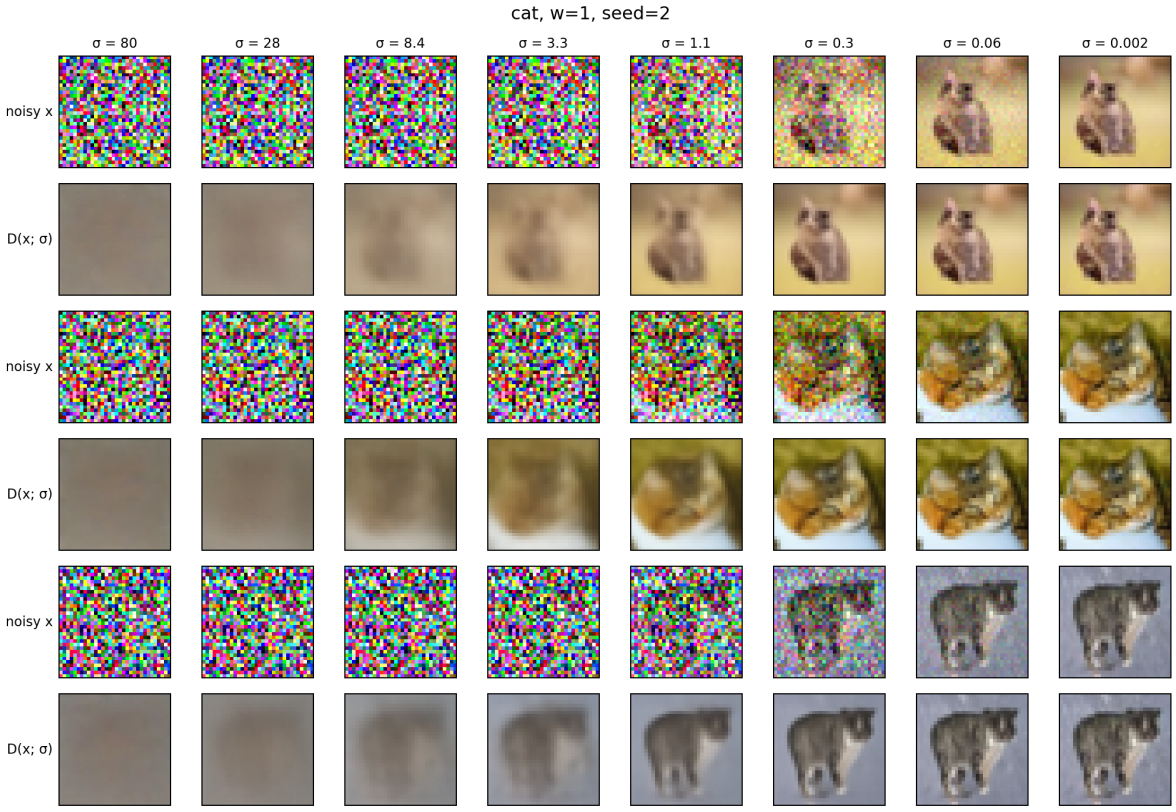


Figure 5: At $w = 1$, cat structure is evident by $\sigma \approx 1.1$ and composition is mostly stable below $\sigma \approx 0.3$.

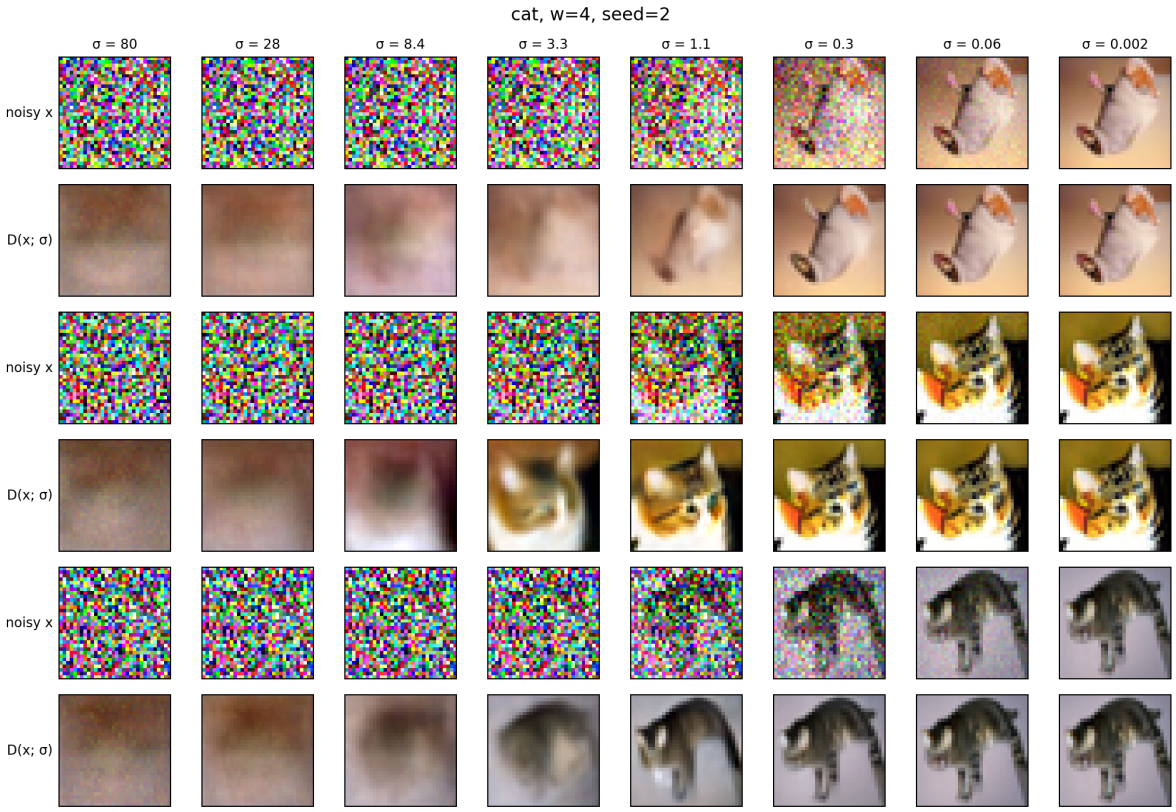


Figure 6: The paired $w = 4$ trajectory shows clearer class structure earlier in the second example and stronger contrast at low noise.

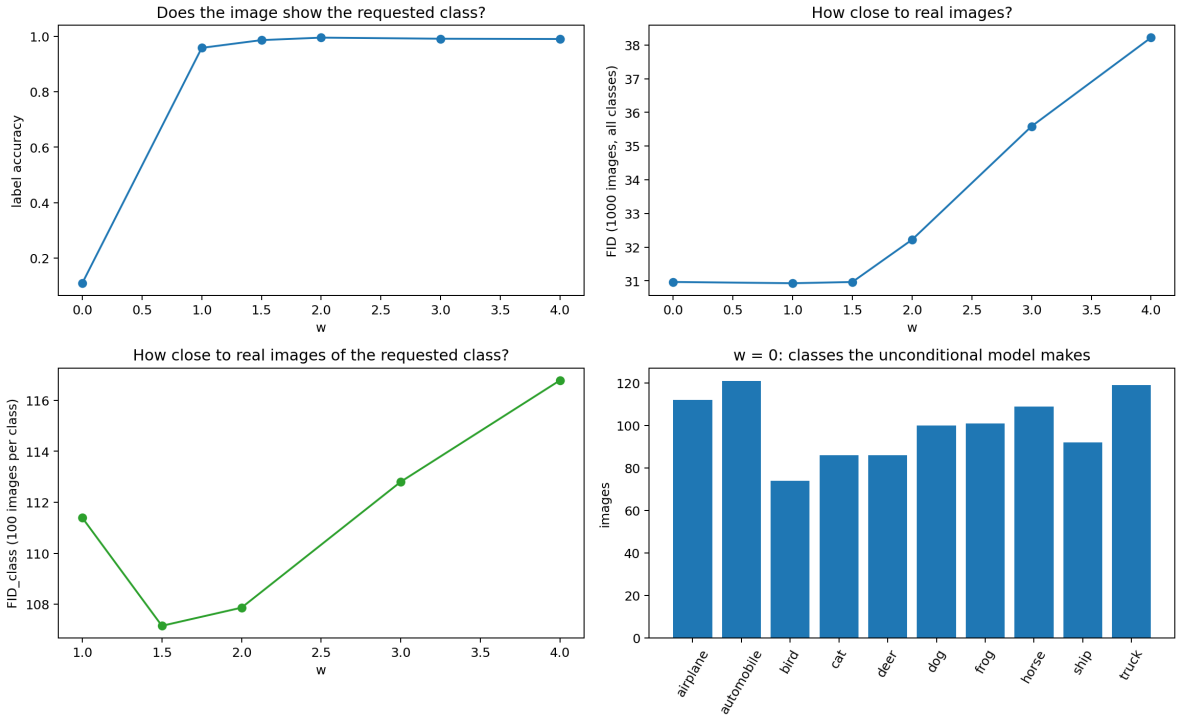


Figure 7: Moderate guidance improves requested-class accuracy; larger weights worsen global and class FID.

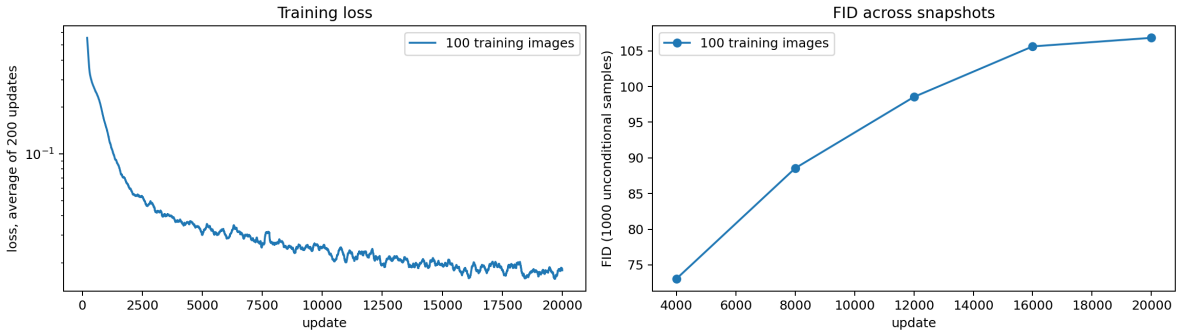


Figure 8: With 100 training images, training loss falls while snapshot FID rises.

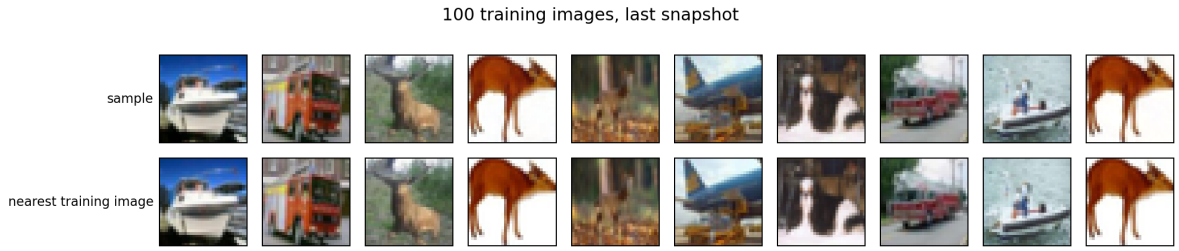


Figure 9: The 100-image model closely matches training-image objects and backgrounds, with much smaller nearest-image distances than real test examples.

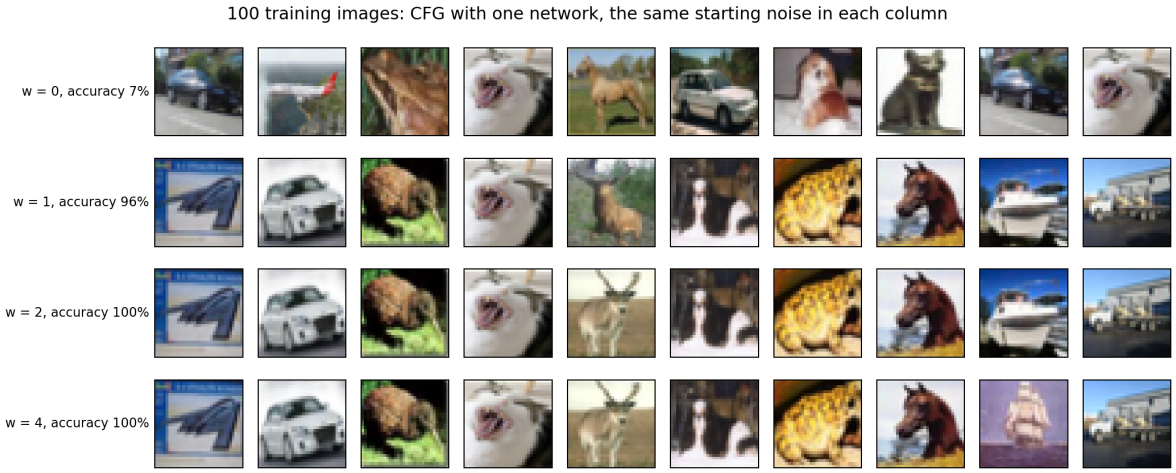


Figure 10: The 100-image model's requested-class accuracy rises from 96% at $w = 1$ to 100% at $w = 2$.

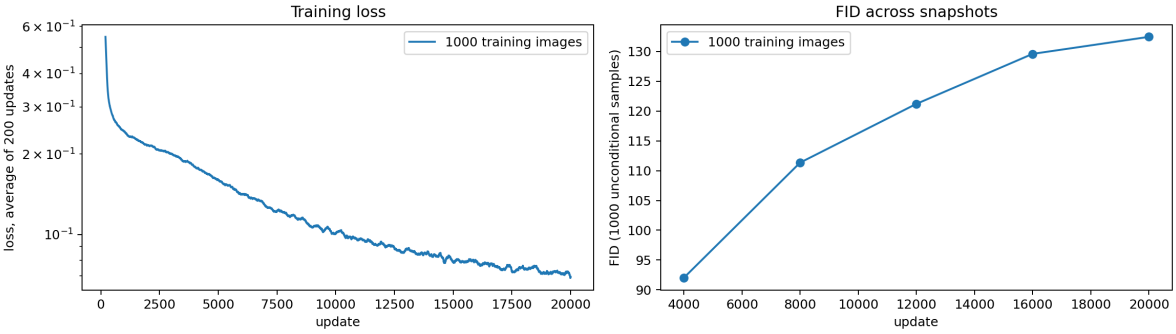


Figure 11: With 1,000 training images, lower loss also accompanies higher snapshot FID.

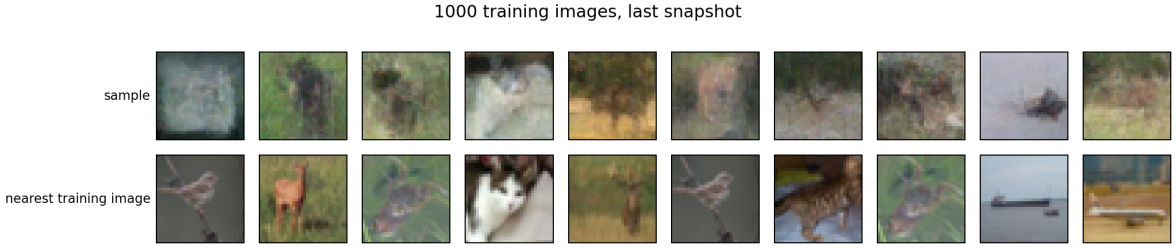


Figure 12: The 1,000-image model shows partial training-image matches and blurred outputs; exact-copy evidence is weaker than for 100 images.

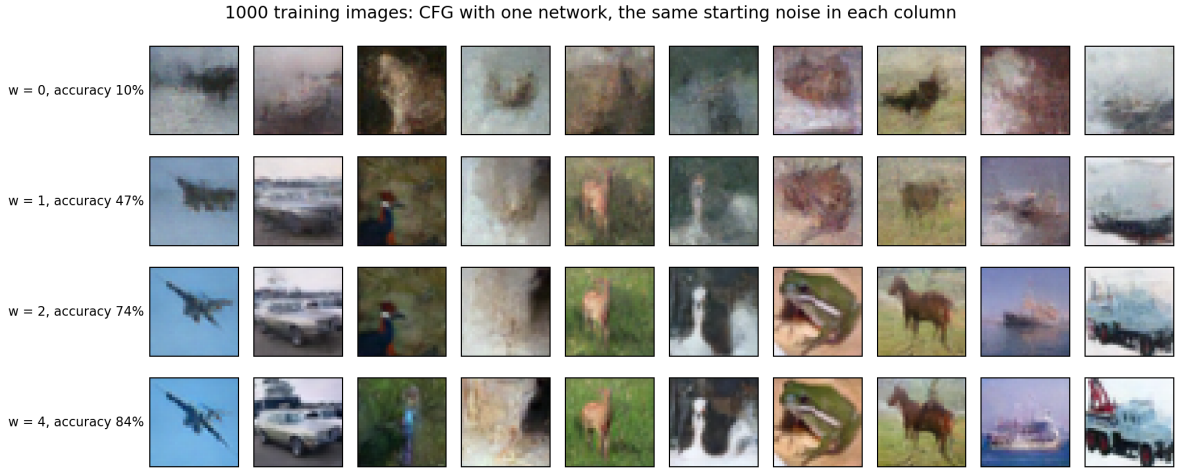


Figure 13: Guidance improves the 1,000-image model's accuracy from 47% at $w = 1$ to 84% at $w = 4$, while artifacts remain.

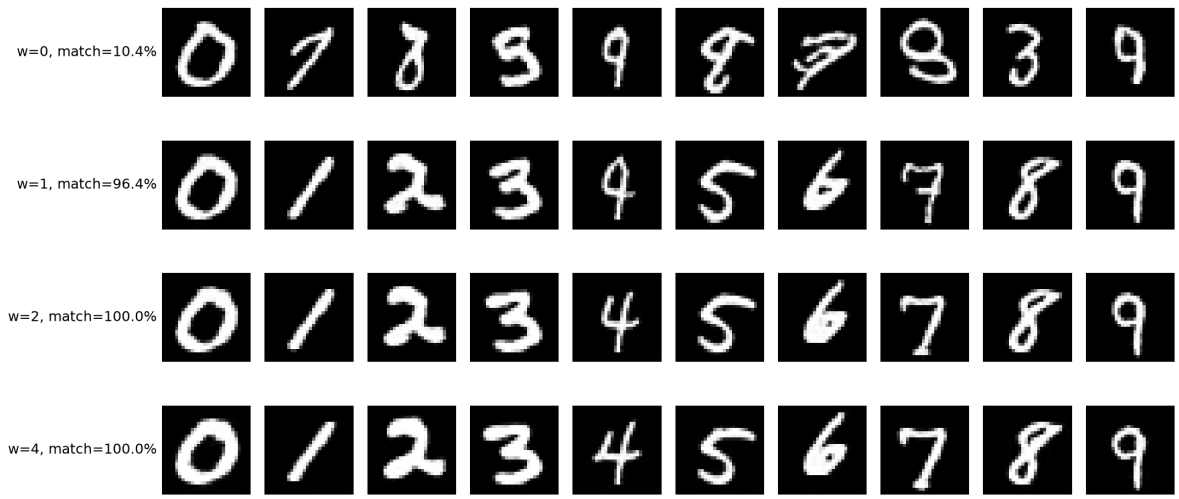


Figure 14: Guidance improves requested-digit matching; the requested seven becomes clearer at higher weights.

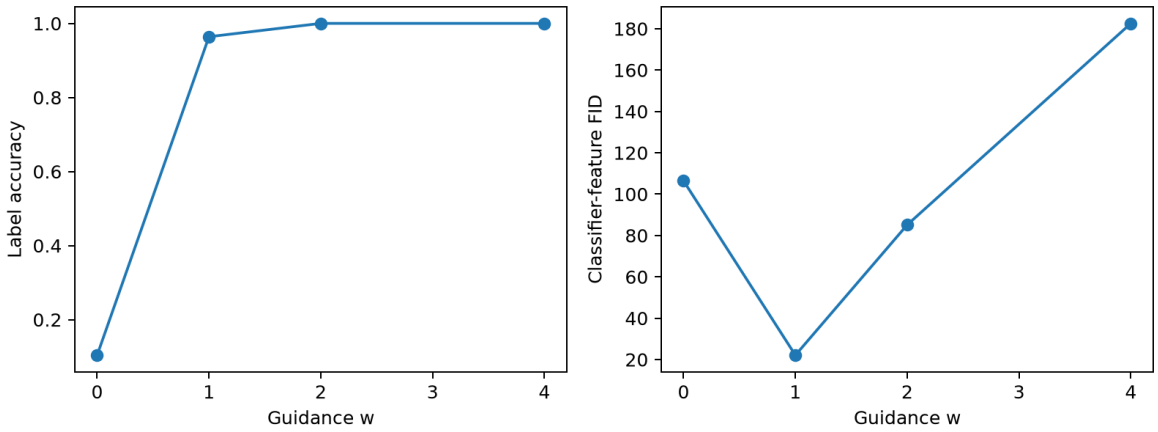


Figure 15: MNIST feature FID is lowest at $w = 1$; $w = 2$ achieves perfect measured accuracy with lower FID than $w = 4$.

Reproducibility and AI use

The public GitHub repository documents data downloads, dependencies, and reproduction commands. Frozen-classifier real-test accuracies were 94.4% on CIFAR-10 and 99.32% on MNIST. The encouraged full-size EDM training run was prepared but not run.

MNIST resumed at update 6,500 with its training-noise stream restarted; all guidance settings use the same final weights and paired evaluation noise. The 100-image CIFAR run transferred from Mac to CUDA at update 1,500; the 1,000-image run used CUDA throughout. Manifests and notes record these details.

I used OpenAI Codex for development, debugging, experiment automation, and writing assistance. My understanding of the methods comes from reading the EDM paper myself.

References

[1] T. Karras, M. Aittala, T. Aila, and S. Laine. Elucidating the Design Space of Diffusion-Based Generative Models. NeurIPS, 2022. https://arxiv.org/abs/2206.00364.

[2] J. Ho, A. Jain, and P. Abbeel. Denoising Diffusion Probabilistic Models. NeurIPS, 2020. https://arxiv.org/abs/2006.11239.

[3] Y. Song, J. Sohl-Dickstein, D. P. Kingma, A. Kumar, S. Ermon, and B. Poole. Score-Based Generative Modeling through Stochastic Differential Equations. ICLR, 2021. https://arxiv.org/abs/2011.13456.

[4] J. Ho and T. Salimans. Classifier-Free Diffusion Guidance. arXiv:2207.12598, 2022. https://arxiv.org/abs/2207.12598.